

Data Mining for Business Analytics

MSBA 511

Text Mining

Introduction to Text Mining

Up until now we have only worked with **structured quantitative data** (e.g., numbers, binary/categorical variables). Now, we will shift our focus to **unstructured text data** such as:

- Tweets (Social media posts)
- Customer reviews
- Free-form response fields
- Medical records

Goal: Transform raw textual information and label document as belonging to a class or group similar text documents.



The broader term natural language processing (NLP) is more commonly used today with an emphasis on deriving understanding of the text and interpreting it like a human.

Why Mine Text Data?

Challenge: Valuable insights often hidden in unstructured claim notes.



CLAIM FORM AND INSTRUCTIONS

If you have any questions regarding benefits available, or how to file your claim, or if you would like to appeal any determination, please contact our Customer Care Center at 1-800-348-4489, 8:00 A.M. to 8:00 P.M. Eastern Standard Time

The furnishing of this form, or its acceptance by the Company as proof, must not be construed as an admission of any liability on the part of the Company, nor a waiver of any of the conditions of the insurance contract.

INSTRUCTIONS FOR FILING YOUR GROUP ACCIDENT CLAIM

Approach:

- Extract and transform text from claim forms into features.
- Train a predictive model using labeled past claims (fraudulent vs. non-fraudulent).
- Apply the model to flag suspicious new claims for review.

Impact: Enhances fraud detection by incorporating qualitative insights from written text.

Why Mine Text Data?

Challenge: Clinics receive a high volume of online appointment requests written in free text.

✓ Department

Reason

Visit type

Location

Date and time

Book

What's the reason for your appointment?

Describe a symptom, condition, or reason for a visit

ENTER REASON (Required)

Example: I've had left knee pain for 3 months.

300 of 300 characters left

Cancel Back Next

Approach:

- Use text mining to extract key symptoms and urgency indicators from patient-submitted forms.
- Classify and route requests to the appropriate level of care:
 - **Administrative Assistant** (e.g., scheduling or insurance questions)
 - **Nurse** (e.g., minor issues or follow-ups)
 - **Doctor** (e.g., urgent or complex medical needs)

Impact: Streamlines triage, reduces delays, and ensures patients receive the right level of care promptly.

Why Mine Text Data?

Challenge: Businesses receive large volumes of unstructured customer reviews.

Approach:

- Use text mining to identify key themes, sentiments, and product/service feedback.
- Highlight what customers liked or disliked, and extract frequently mentioned features.
- Summarize or highlight helpful reviews to assist future buyers in decision-making

Impact: Enhances customer experience, supports product improvement, and aids potential customers with relevant **insights**.

Customer reviews

★★★★☆ 4.6 out of 5

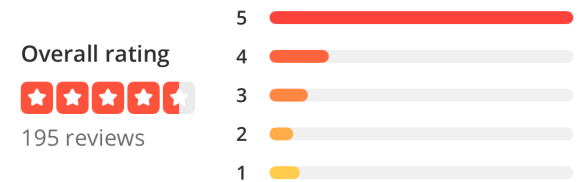
5,006 global ratings

Customers say

Customers find the book excellent for developing data visualization strategies, with practical examples and easy-to-follow explanations. Moreover, the visual quality receives positive feedback for its illustrations on real examples, and customers appreciate how it delivers compelling data-driven stories while making the topic fun to read. Additionally, the book effectively eliminates clutter in charts. However, opinions about the book's content are mixed, with some finding it opinionated, and customers disagree on its value for money.

AI-generated from the text of customer reviews

Recommended reviews



Q Search

Yelp sort

Filter by rating

Insights

😊 Ambience (45)

😊 Drinks (8)

😊 Food (110)

😊 Service (71)

😊 Wait time (6)

Key Steps in Text Mining

1. Text Collection

Gather unstructured data from sources like social media, emails, reports, reviews, etc.

2. Text Preprocessing

- Tokenization: Split text into words or terms
- Ignore case, stop word removal, lemmatization/stemming
- Noise removal: punctuation, numbers, special characters

3. Text Representation

Convert text into numerical format using:

- Bag-of-Words
- TF-IDF (Term Frequency-Inverse Document Frequency)

4. Modeling and Analysis

Apply data mining techniques such as clustering and classification.

Bag-of-Words

The term Bag-of-Words (BoW) comes from the fact it treats text just as a collection (**bag**) of individual words:

- Ignores grammar, syntax, punctuation, and word order.
- Focuses on word frequency or presence, not structure

This technique is surprisingly effective for tasks like:

- Classification tasks especially for short-form text
- Sentiment identification for content such as customer reviews

Common in supervised learning with large text **corpora**.

- ✓ Does not require perfect accuracy to deliver valuable insights.



Corpora often refers to a fixed standard set of text documents that many researches can use to develop and tune text mining algorithms.

Tokenization

The first step is to break the text up into individual terms. Tokenization is the process of identifying delimiters (e.g., spaces, punctuation) and separate the text into meaningful units called **tokens**.

For large text collections, tokenization can produce thousands to millions of unique terms. The English language alone has over 1 million words, not including misspellings, slang, or symbols.

Challenge:

High dimensionality can hinder analysis and model performance.

Solution:

Apply preprocessing techniques to reduce the number of terms and focus on the most informative tokens.

Lemmatization vs. Stemming

Lemmatization and stemming are both techniques used to reduce words to their root form but differ in accuracy and linguistic awareness.

Feature	Stemming	Lemmatization
Definition	Remove suffixes to reduce words to base/root form	Convert words to their dictionary (lemma) form
Example	"running" → "run" "technical" → "technic"	"running" → "run" "better" → "good"
Method	Rule-based truncation	Dictionary and grammar-based lookup
Speed	Fast, simple	Slower, more accurate
Output Quality	May create non-words	Produces valid words
Use Case	Large-scale, quick preprocessing	NLP tasks needing higher linguistic accuracy

Text Preprocessing Example

Let's practice and work through a simple text example:

The technician was resolving technical issues quickly.

Step 1: Tokenization based on the space delimiter:

["The", "technician", "was", "resolving", "technical", "issues", "quickly"]

Step 2: Ignore case and stop word removal

["technician", "resolving", "technical", "issues", "quickly"]

Step 3: Stemming

["technician", "resolv", "technic", "issu", "quick"]

BoW Matrix Representation

- Columns are terms
- Rows are text/documents
- Cells indicate binary or frequency of terms in documents

Let's review the matrix representation for the following:

S1: The technician was resolving technical issues quickly.

S2: The engineer resolved several technical problems efficiently.

Text	technician	resolv	technic	issu	quick	engin	sever	problem	effici
S1	1	1	1	1	1	0	0	0	0
S2	0	1	1	0	0	1	1	1	1



There are many different stemming algorithms (Porter, Snowball etc.), and they can produce different stems for the same word. Always choose a method that balances simplicity, accuracy, and consistency for your analysis.

Real Text is Not that Simple

Numbers (including dates, percents, monetary amounts), e.g., from Google Annual Report 2014:

We considered the historical trends in currency exchange rates and determined that it was reasonably possible that changes in exchange rates of 20% could be experienced in the near term. If the U.S. dollar weakened by 20% at December 31, 2013 and 2014, the amount recorded in AOCI related to our foreign exchange options before tax effect would have been approximately \$4 million and \$686 million lower at December 31, 2013 and December 31, 2014, and the total amount of expense recorded as interest and other income, net, would have been approximately \$123 million and \$90 million higher in the years ended December 31, 2013 and December 31, 2014. If the U.S. dollar strengthened by 20% at December 31, 2013 and December 31, 2014, the amount recorded in accumulated AOCI related to our foreign exchange options before tax effect would have been approximately \$1.7 billion and \$2.5 billion higher at December 31, 2013 and December 31, 2014, and the total amount of expense recorded as interest and other income, net, would have been approximately \$120 million and \$164 million higher in the years ended December 31, 2013 and December 31, 2014.

Real Text is Not that Simple

Email addresses, URL's, stray characters:

Sender: Distribution list for statistical items of interest <WSS-ELECTRONIC-MAIL-LIST@LISTS.MITRE.ORG> From: "Massimini, Vince" <svm@mitre.org> Subject: Comparing the Maximal Procedure to Permuted Blocks Randomization To: <WSS-ELECTRONIC-MAIL-LIST@LISTS.MITRE.ORG> Precedence: list List-Help: <mailto:LISTSERV@LISTS.MITRE.ORG?body=INFO%20WSS-ELECTRONIC-MAIL-LIST>

For more WSS events, see washstat.org WSS Public Health/Biostatistics Section and NCI Division of Cancer Prevention on Jointly Sponsored Event: =20

SPEAKER: Vance W. Berger, PhD National Cancer Institute and University of Maryland Baltimore County and Klejda Bejleri, BS Biometry and Statistics, Department of Biological Statistics and Computational Biology, Cornell University, Ithaca, NY 14853 =20 TITLE: Comparing the Maximal Procedure to Permuted Blocks Randomization

TIME AND PLACE: Monday, June 8th NCI Shady Grove, 9609 Medical Center Drive, Rockville MD Room 5E30/32. Bring photo ID, allow time to get through security

Real Text is Not that Simple

Proper nouns and terms specific to a specific field:

From Techsmith corporate information: All-In-One Capture, Camtasia, Camtasia Studio, Camtasia Relay, Coach's Eye, DubIt, EnSharpen, Enterprise Wide, Expressshow, Jing, Morae, Rich Recording Technology (RRT), Snagit, Screencast.com, ScreenChomp, Show The World, SmartFocus, TechSmith, TechSmith and T Design logo, TechSmith Fuse, TechSmith Relay, TSCC, and UserVue are marks or registered marks of TechSmith Corporation.

From medical journal: Eight hundred elderly women and men from the population-based Framingham Osteoporosis Study had BMD assessed in 1988-1989 and again in 1992-1993. BMD was measured at femoral neck, trochanter, Ward's area, radial shaft, ultradistal radius, and lumbar spine using Lunar densitometers. (Risk Factors for Longitudinal Bone Loss in Elderly Men and Women: *The Framingham Osteoporosis Study*, *Journal of Bone and Mineral Research* [Volume 15, Issue 4](#), pages 710–720, April 2000)

TF-IDF Representation

Another popular option that factors in both the frequency of a term in the document as well as all documents with that term is **Term Frequency-Inverse Document Frequency (TF-IDF)**.

$TF(t, d)$ = Number of times t appears in document d

The **Inverse Document Frequency (IDF)** for term t is calculated across all documents and defined as:

$$IDF(t) = 1 + \log\left(\frac{\text{Total number of documents}}{\text{Number of documents with } t}\right)$$

Finally, the **TF-IDF(t, d)** for a specific term-document pair is:

$$TF - IDF(t, d) = TF(t, d) \times IDF(t)$$

Let's Practice: TF-IDF

S1: The technician was resolving technical issues quickly.

S2: The engineer resolved several technical problems efficiently.

Text	technician	resolv	technic	issu	quick	engin	sever	problem	effici
S1	1	1	1	1	1	0	0	0	0
S2	0	1	1	0	0	1	1	1	1

$$TF(\text{technician}, S1) = 1 \quad TF(\text{technician}, S2) = 0$$

$$IDF(\text{technician}) = 1 + \log\left(\frac{2}{1}\right) = 1.69$$

$$TF - IDF(\text{technician}, S1) = 1 \times 1.69 = 1.69$$

$$TF - IDF(\text{technician}, S2) = 0 \times 1.69 = 0$$

Let's Practice: TF-IDF

S1: The technician was resolving technical issues quickly.

S2: The engineer resolved several technical problems efficiently.

Text	technician	resolv	technic	issu	quick	engin	sever	problem	effici
S1	1	1	1	1	1	0	0	0	0
S2	0	1	1	0	0	1	1	1	1

$$TF(resolv, S1) = 1 \qquad TF(resolv, S2) = 1$$

$$IDF(resolv) = 1 + \log\left(\frac{2}{2}\right) = 1.00$$

$$TF - IDF(resolv, S1) = 1 \times 1.00 = 1.00$$

$$TF - IDF(resolv, S2) = 1 \times 1.00 = 1.00$$

Bag-of-Words vs. TF-IDF

Use Bag-of-Words when frequent terms matter (e.g., spam, sentiment clues) and TF-IDF when differentiation of text is important.

Criteria	Bag-of-Words	TF-IDF
Best For	Detecting patterns on frequent words	Highlighting unique or distinguishing words
Best Use Cases	Spam Detection, Sentiment Classification	Document similarity, Topic Modeling, Content Recommendations
Frequent Words	Tread as strong signal	Down weighted
Rare Words	May have little impact	Get higher weights and can be useful for differentiation
Short Texts	Often more effective due to limited context	May under weight rare but important terms
Long Texts	May be overwhelmed by common words	Handles length and vocabulary variation better
Interpretability	Simple and intuitive	Requires understanding of log-weighting

Terms to Concepts

- The post-reduction term/document matrix is often still huge – too big for easy processing.
- Recall how, with principal components, we derived a small set of synthetic predictor variables, each of which was a linear combination of “like-minded” original variables.
- Latent semantic indexing does something similar for text – it maps multiple terms to a small set of **concepts**.

Extracting Meaning

- It may be possible to use the concepts to identify themes in the document corpus, and clusters of documents sharing those themes.
- Often, however, the concepts do not map in obvious fashion to meaningful themes.
- The key contribution is simply reducing the vocabulary – instead of a matrix with thousands of columns, we can deal with just a small subset.